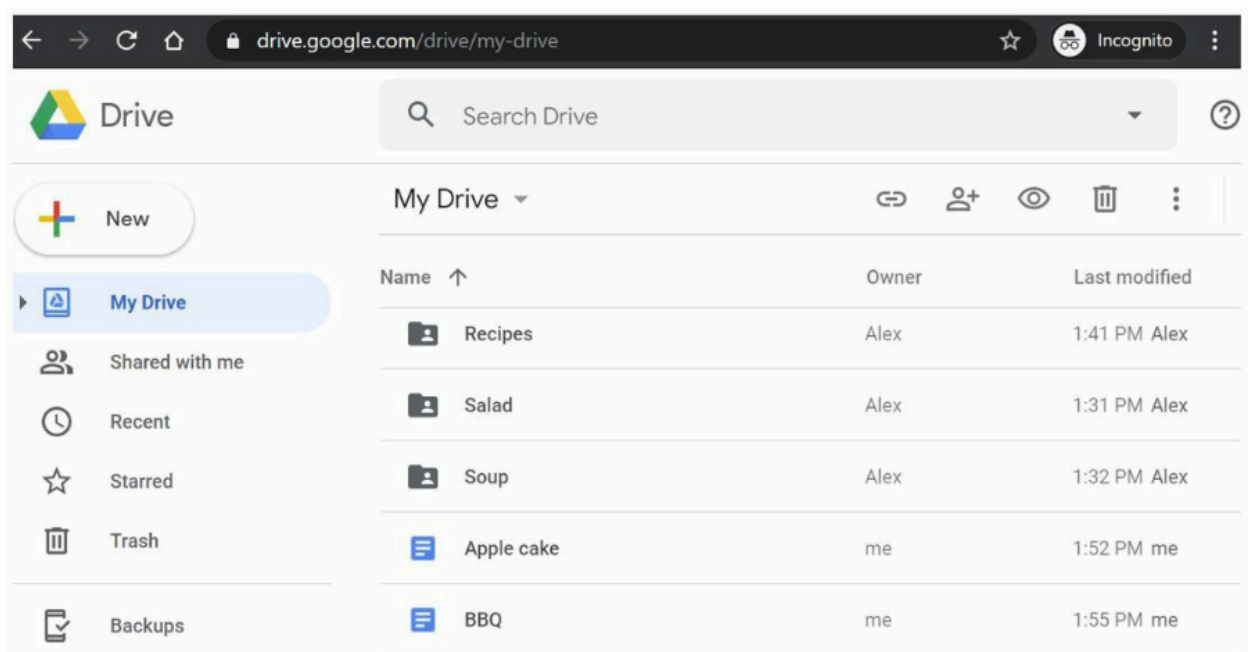
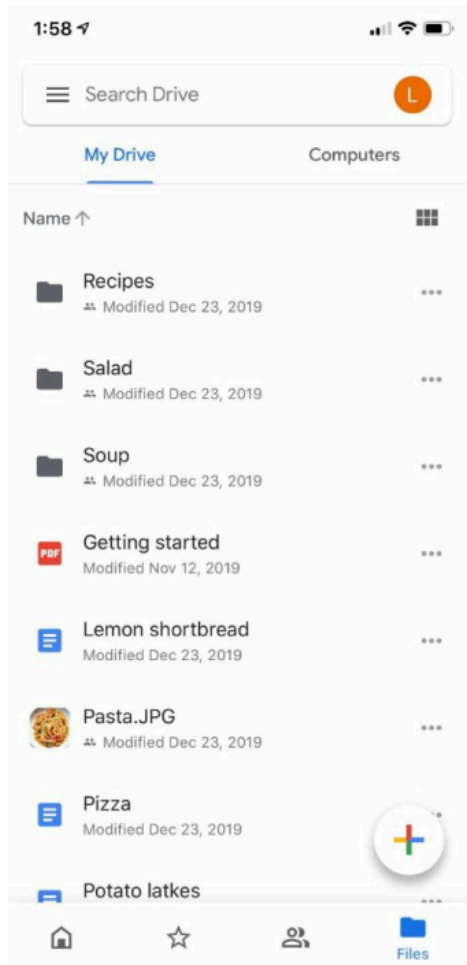


Design Google Drive

In recent years, cloud storage services such as [Google Drive](#), [Dropbox](#), [Microsoft OneDrive](#), and [Apple iCloud](#) have become very popular.

Let us take a moment to understand Google Drive before jumping into the design. Google Drive is a file storage and synchronization service that helps users store documents, photos, videos, and other files in the cloud. Users can access files from computers, smartphones, and tablets. The platform also supports file sharing and synchronization across multiple devices.





Core Features of Google Drive

- File upload and download
- File synchronization across devices
- File sharing and collaboration
- Cloud backup and recovery
- Real-time access from anywhere
- Version history management

Functional Requirements

- Users should be able to upload files.
- Users should be able to download files.
- Files should synchronize automatically across multiple devices.

- Users should be able to share files with others.
- The system should support large file storage.
- The system should maintain file versions and recovery options.

Non-Functional Requirements

- High availability
- Scalability
- Reliability and durability
- Fast synchronization speed
- Low latency
- Strong security and privacy protection

Important Design Goals

- Efficient file synchronization
- Reliable distributed storage
- Optimized bandwidth usage
- Fault tolerance
- Secure file sharing and permissions
- Support for billions of files and millions of users